Supplementary Material for “Battery Dispatch Optimization Considering Cell Voltage Profiles: A Piecewise Input Convex Neural Network Approach”

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I. Proof of Theorem

With given $W^t_{i,j,k}$, $V^t_{i,j,k}$, and $b^t_{i,j,k}$ in (1a) for the trained PICNN, the bilinear term can be accurately relaxed using the McCormick linearization method, as each ReLU activation function is inherently continuous:

$$z^t_{i,j,k} \leqslant \sigma((x^t_i)^{\mathrm{T}} W^t_{i,0,k} + b^t_{i,0,k}) + (1-\alpha^t_k)M, \quad \text{if } j=1, \tag{A-1a}$$

$$z^t_{i,j,k} \geqslant \sigma((x^t_i)^{\mathrm{T}} W^t_{i,0,k} + b^t_{i,0,k}) + (\alpha^t_k - 1)M, \quad \text{if } j=1, \tag{A-1b}$$

$$\begin{aligned} z^t_{i,j,k} \leqslant\, & \sigma((z^t_{i,j-1,k})^{\mathrm{T}} W^t_{i,j-1,k} + (x^t_i)^{\mathrm{T}} V^t_{i,j-1,k} + b^t_{i,j-1,k}), \\ & +(1-\alpha^t_k)M, \qquad \text{if } j=2,\ldots,N_I-1, \end{aligned} \tag{A-1c}$$

$$\begin{aligned} z^t_{i,j,k} \geqslant\, & \sigma((z^t_{i,j-1,k})^{\mathrm{T}} W^t_{i,j-1,k} + (x^t_i)^{\mathrm{T}} V^t_{i,j-1,k} + b^t_{i,j-1,k}), \\ & +(\alpha^t_k - 1)M, \qquad \text{if } j=2,\ldots,N_I-1, \end{aligned} \tag{A-1d}$$

$$z^t_{i,j,k} = (z^t_{i,j-1,k})^{\mathrm{T}} W^t_{i,j-1,k} + (x^t_i)^{\mathrm{T}} V^t_{i,j-1,k} + b^t_{i,j-1,k}, \ \text{if } j=N_I, \tag{A-1e}$$

where M is the given large positive scalar.

Then, since each output $z^t_{i,j}$ is convex with respect to the input vector x^t_i for the k-th range of variable x^t_i, i.e., $\alpha^t_k = 1$, if all linear-term coefficients are nonnegative, and the activation function $\sigma(\cdot) = \max(\cdot, 0)$ is convex and non-decreasing. That yields that (A-1a) and (A-1c) can be negligible, and thus (A-1a)–(A-1e) can be equivalently formulated as

$$\min_{x^t_i, z^t_{i,j,k}, \alpha^t_k} \sum_{k \in \mathcal{K}} z^t_{i,N_I,k}, \tag{A-2a}$$

$$z^t_{i,j,k} \geqslant (x^t_i)^{\mathrm{T}} W^t_{i,0,k} + b^t_{i,0,k} + (\alpha^t_k - 1)M, \quad \text{if } j=1, \tag{A-2b}$$

$$\begin{aligned} z^t_{i,j,k} \geqslant\, & (z^t_{i,j-1,k})^{\mathrm{T}} W^t_{i,j-1,k} + (x^t_i)^{\mathrm{T}} V^t_{i,j-1,k} + b^t_{i,j-1,k}, \\ & +(\alpha^t_k - 1)M, \qquad \text{if } j=2,\ldots,N_I-1, \end{aligned} \tag{A-2c}$$

$$z^t_{i,j,k} = (z^t_{i,j-1,k})^{\mathrm{T}} W^t_{i,j-1,k} + (x^t_i)^{\mathrm{T}} V^t_{i,j-1,k} + b^t_{i,j-1,k}, \ \text{if } j=N_I. \tag{A-2d}$$

Clearly, this model (A-2a)–(A-2d) is the same with the MILP-based model (2a)–(2d), which proves the Theorem. ■

II. Charging Current Dynamics under CCCV Mode

When charging the battery cell in CCCV mode, the charging current surface can be exhibited in Fig. 1. In this figure, $I^t_{\mathrm{D},i}$ denotes the variable charging current during Δt shown in the coloured surface once $U^t_{\mathrm{c,i}} = \overline{U}_{\mathrm{c}}$ holds, and the grey-coloured plane denotes $I^t_{\mathrm{L},i} = I^t_{\mathrm{D},i}$ if $U^t_{\mathrm{c,i}} < \overline{U}_{\mathrm{c}}$ satisfies. These two areas are divided by the cut line $\overline{ab}$.

It is clear that the sharp dynamics of charging current via a CCCV mode results in nonlinear variations of $I^t_{\mathrm{D},i}$. Due to $\Delta S^t_i = \frac{1}{B_i}\int_0^{\Delta t} I^t_{\mathrm{D},i} U^t_{\mathrm{c},i}\,\mathrm{dt}$, then ΔS^t_i will change in a nonlinear manner. However, according to the classic battery SoC equation in [4], we have

$$\Delta S^t_i = \eta_{c,i} p^t_{c,i} \Delta t + \eta_{d,i} p^t_{d,i} \Delta t, \tag{A-3}$$

where $\eta_{c,i}$ and $\eta_{d,i}$ respectively denote the fixed charging and discharging efficiency coefficients for the i-th battery; and

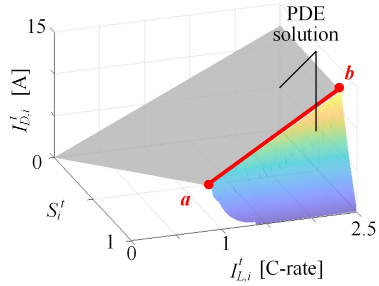

Fig. 1. Current surface of charging a battery cell in CCCV mode.

$p^t_{c,i}$ and $p^t_{d,i}$ respectively denote the charging and discharging power.

For Δt=1 hour, we can quantify $\eta_{c,i}$ and $\eta_{d,i}$ via COMSOL simulation under CCCV charging and CC discharging cycles in Fig. 2(a) and (b), respectively.

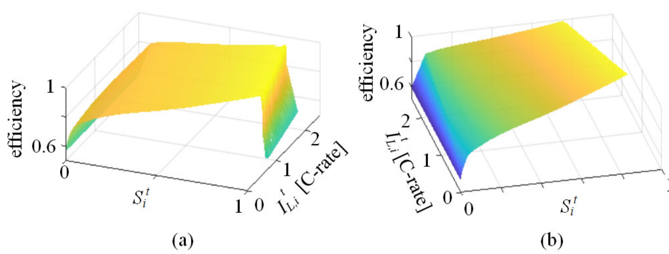

Fig. 2. Efficiency coefficients: (a) $\eta_{c,i}$ under CCCV charging; and (b) $\eta_{d,i}$ under CC discharging.

As observed in Fig. 2(a) and (b), charging and discharging efficiency coefficients $\eta_{c,i}$ and $\eta_{d,i}$ vary in the ranges $[0.6, 0.98]$ and $[0.6, 0.99]$, which are functions of $(S^t_i, I^t_{\mathrm{L},i})$. This means that the classic battery SoC equation with fixed efficiency coefficients lead to the inaccuracy of BDO solutions. However, with the ICNN-based ΔS^t_i learned from PDE-based simulation data, the SoC equations can be made both more accurate and simpler, also eliminating the need for charging and discharging power variables.

III. Comparing PICNN with ICNN for Accuracy

We compare the proposed multiple ICNN-based approximation with the conventional ICNN approach [7] for the solution's accuracy. Figs. 3(a) and (b) respectively plot the voltage surfaces of charging a battery cell in CCCV mode and discharging in CC mode using the conventional ICNN approach. The conventional ICNN consists of $N_I = 4$ layers with dimensions of 16, 8, 4, and 1, respectively, with initial learning rate 0.02 decaying by 0.005 over 1200 iterations. It is trained using a total of 2,500 sample data points generated by COMSOL simulations.

TABLE I
PARAMETERS FOR ELECTROCHEMICAL KINETICS

Parameter	Negative electrode	Separator	Positive electrode
electrode plate area (m^2)	0.163	0.163	0.163
electrode thickness (m)	$78 \cdot 10^{-6}$	$20 \cdot 10^{-6}$	$45 \cdot 10^{-6}$
Li^+ diffusion coefficient (m^2/s)	$3.9 \cdot 10^{-5}$	-	$1.8 \cdot 10^{-8}$
active electrode volume fraction (%)	0.6	-	0.6
electrolyte phase volume fraction (%)	0.3	-	0.3
max solid phase concentration (mol/m^3)	31507	-	49000
particle radius (m)	$6 \cdot 10^{-6}$	-	$5 \cdot 10^{-6}$
reaction rate efficiency (A/m^2)	$9.77 \cdot 10^{-2}$	-	$1.19 \cdot 10^{-2}$
exchange current density of side reaction (A/m^2)	10	-	10
initial electrolyte concentration (mol/m^3)	$1.25 \cdot 10^4$	$1.25 \cdot 10^4$	$1.25 \cdot 10^4$
Binder volume fraction (%)	0.1	-	0.1
Separator volume fraction (%)	-	0.4	-

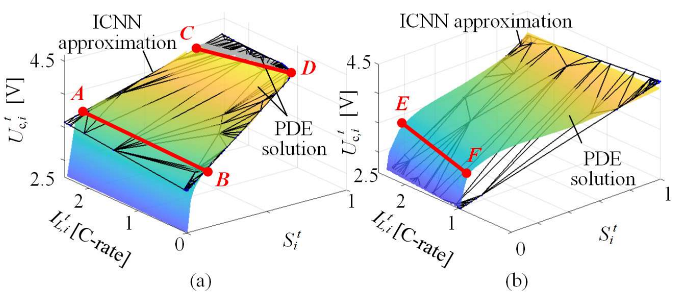

Fig. 3. Voltage surfaces of a battery cell: (a) charging in CCCV mode; and (b) discharging in CC mode.

As compared to Fig.1(a) and (b), the approximation error can be obviously larger in Figs. 3(a) and (b), since cell voltage surfaces are significantly influenced by polarization effects in CCCV/CC modes. Followed by PWL and domain knowledge from battery physics, employing multiple ICNN-based approximation functions for different SoC ranges achieves higher accuracy: the maximum error between the PDE-based solution and the proposed multi-ICNN approach is less than 2%, compared to 11% in Fig. 3(a) and 67% in Fig. 3(b) for a single ICNN-based approximation.

IV. DERIVATION OF SECOND-ORDER CONIC CONSTRAINT

Let $\mathcal{F} = (z^t_{i,N_I,k})^2$, and then the epigraph constraint is expressed as

$$\text{epi } \mathcal{F} := \{(\gamma^t_{i,k}, z^t_{i,N_I,k}) | \gamma^t_{i,k} \geqslant \mathcal{F}(z^t_{i,N_I,k})\} \tag{A-4}$$

This epigraph constraint can be derived as a second-order conic constraint. Define $f := (\gamma^t_{i,k}+1)^2$ and $g := (\gamma^t_{i,k}-1)^2 + (2z^t_{i,N_I,k})^2$. Then

$$f - g = 4\gamma^t_{i,k} - 4(z^t_{i,N_I,k})^2 \geqslant 0, \tag{A-5}$$

where the inequality holds because $\gamma^t_{i,k} \geqslant (z^t_{i,N_I,k})^2$. Rearrangement of (A-5) yields

$$(\gamma^t_{i,k}+1)^2 \geqslant (\gamma^t_{i,k}-1)^2 + (2z^t_{i,N_I,k})^2, \tag{A-6}$$

which is equivalent to the standard second-order conic constraint

$$\left\|\gamma^t_{i,k}+1\right\|_2^2 \geqslant \left\|[\gamma^t_{i,k}-1,\ 2z^t_{i,N_I,k}]^{\mathrm{T}}\right\|_2^2. \tag{A-7}$$

V. BATTERY PARAMETER VALUES

We select the total NCM battery capacity $B_i = 200\,\text{Ah}$ for unmanned electric ships (UES) with the operating voltage ranging from $2.6\,\text{V}$ to $4.2\,\text{V}$, and the cell current $I^t_{\text{L},i} \in [0, \overline{I}_{\text{L},i}]$ and $\overline{I}_{\text{L},i} = 2.5\,\text{C-rate}$. Other parameters of a NCM battery cell for electrochemical kinetics in COMSOL simulation are given in Table I.

The cell voltage profile constraint in M_1 [6] is formulated as

$$(U^t_{\text{c},i})^2 \leqslant \overline{U}_{\text{c}} U^t_{\text{c},i} - R_i p^t_{c,i} \tag{A-8}$$

As indicated in Fig. 4, the grey hyperplane represents the fixed cell voltage profile, which exhibits significant approximation error for the PDE solution by COMSOL especially when S^t_i approaches 0 or 1. Using M_1, the fixed resistance R_i for a battery cell can be fitted as 9.6 p.u.

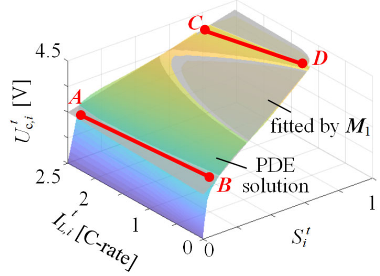

Fig. 4. Fitting the cell voltage profile using M_1 [6].